

CS1231S

TUTORIAL #5

Relations & Partial orders

Adapted from Prof Aaron's slides

Learning objectives of this tutorial

Partial orders

- Determining whether a relation is **antisymmetric**
- Determining whether a relation is **asymmetric**
- Reasoning about **partial orders**
- Drawing **Hasse diagrams** of partial orders
- Knowing about **comparability** and **compatibility** of elements in a partial order
- Distinguishing **minimal/maximal** elements from **smallest/largest** elements
- Understanding **linearizations**

Q1. Let S be the set of all strings over the alphabet $\mathcal{A} = \{s, u\}$.
Define a relation R on S by the following:

$$\forall a, b \in S \ (aRb \Leftrightarrow \text{len}(a) \leq \text{len}(b))$$

where $\text{len}(x)$ denotes the length of x . Is R a partial order?

No, it is not a partial order. Why?

It is not antisymmetric.

Counterexample?

Let R be a relation on a set A .

R is **antisymmetric** iff $\forall x, y \in A \ (x R y \wedge y R x \Rightarrow x = y)$.

Q2.

[AY2022/23 Semester 1 Midterm Test]

Let $A = \{2, 3, 5, 7, 21, 30, 84, 99\}$ and let $\preceq$ be a partial order on the set A defined by the “divides” relation, that is, $x \preceq y \Leftrightarrow x|y$. Which of the following statements are true?

- ✗ (i) The partial order has a linearization $\preceq^*$ such that $21 \preceq^* 7$.
- ✓ (ii) The partial order has a linearization $\preceq^*$ such that $3 \preceq^* 2$.
- ✓ (iii) The partial order has a linearization $\preceq^*$ such that $21 \preceq^* 5 \preceq^* 84$.
- ✓ (iv) The partial order has a linearization $\preceq^*$ such that $99 \preceq^* 84 \preceq^* 30$.

Noncomparable elements:

- 2 and 3
- 5 and 21
- 30, 84 and 99

Q3. [AY2023/24 Semester 1 Midterm Test]
 Let $A = \{11, 12, 13, 14, 15, 16\}$. For each $x \in A$, define $F_x = \{k \in \mathbb{Z}^+ : k|x\}$, where $|$ is the “divides” relation. Define also a partial order $\preceq$ on A by setting for all $x, z \in A$:

$$x \preceq z \iff (F_x = F_z) \vee (|F_x| < |F_z|).$$

What are the **minimal**, **smallest**, **maximal**, and **largest elements** of A w.r.t. $\preceq$?

Minimal: 11, 13

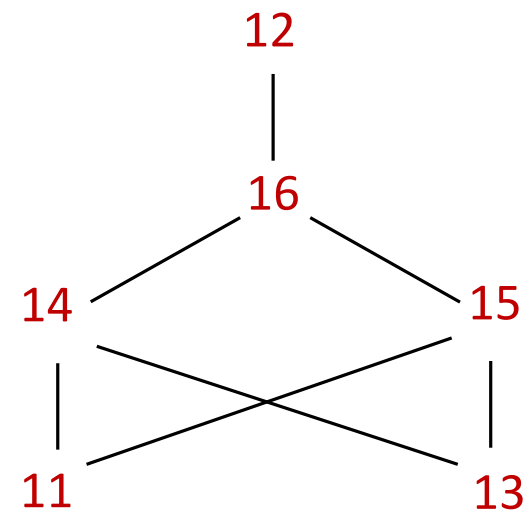
Smallest: None

Maximal: 12

Largest: 12

Let a set A be partially ordered with respect to a relation $\preceq$ and $c \in A$.

1. c is a **maximal element** of A iff $\forall x \in A (c \preceq x \Rightarrow c = x)$.
2. c is a **minimal element** of A iff $\forall x \in A (x \preceq c \Rightarrow c = x)$.
3. c is the **largest element** of A iff $\forall x \in A (x \preceq c)$.
4. c is the **smallest element** of A iff $\forall x \in A (c \preceq x)$.



Q4. Given the partial order $\preceq$ on A in question 3 above, one of the linearizations $\preceq^*$ of $\preceq$ is shown below:

$$11 \preceq^* 13 \preceq^* 14 \preceq^* 15 \preceq^* 16 \preceq^* 12$$

Write out all the other possible linearizations $\preceq^*$.

$$11 \preceq^* 13 \preceq^* 15 \preceq^* 14 \preceq^* 16 \preceq^* 12$$

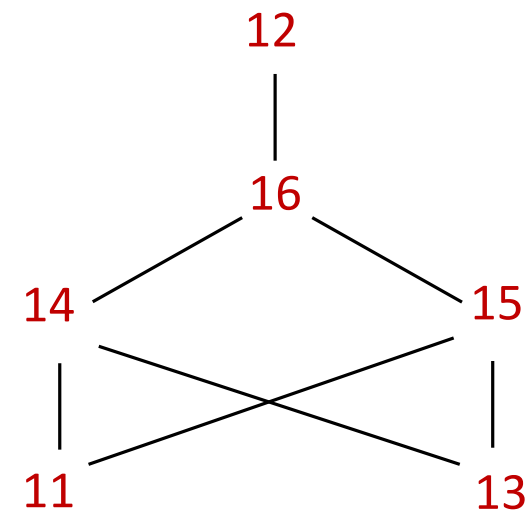
$$13 \preceq^* 11 \preceq^* 14 \preceq^* 15 \preceq^* 16 \preceq^* 12$$

$$13 \preceq^* 11 \preceq^* 15 \preceq^* 14 \preceq^* 16 \preceq^* 12$$

A *linearization* of a partial order $\preceq$ on a set A is a total order $\preceq^*$ on A such that

$$\forall x, y \in A (x \preceq y \Rightarrow x \preceq^* y).$$

Kahn's Algorithm. Pick out a minimal element and place it at the bottom of the total order. Repeat until nothing is left.



Q5. Let A be a set and $\mathcal{P}(A)$ the power set of A .
Prove that the binary relation $\subseteq$ on $\mathcal{P}(A)$ is a partial order.



1. (Reflexivity) Take any $S \in \mathcal{P}(A)$,
 - 1.1. $S \subseteq S$ by the definition of subset.
 - 1.2. Hence $\subseteq$ is reflexive.
2. (Antisymmetry) Take any $S, T \in \mathcal{P}(A)$,
 - 2.1. Suppose $S \subseteq T$ and $T \subseteq S$.
 - 2.2. Then $S = T$ by the definition of set equality.
 - 2.3. Hence $\subseteq$ is antisymmetric.
3. (Transitivity)
 - 3.1. $\subseteq$ is transitive by Theorem 6.2.1.
4. Therefore $\subseteq$ on $\mathcal{P}(A)$ is a partial order.

Theorem 6.2.1.

For all sets A, B, C ,
 $A \subseteq B \wedge B \subseteq C \Rightarrow A \subseteq C$.

Let R be a relation on a set A .

1. R is **reflexive** iff $\forall x \in A (xRx)$.
2. R is **antisymmetric** iff $\forall x, y \in A (xRy \wedge yRx \Rightarrow x = y)$.
3. R is **transitive** iff $\forall x, y, z \in A (xRy \wedge yRz \Rightarrow xRz)$.

Q6. Let $B = \{0,1\}$ and define the binary relation R on $B \times B$ as follows:
 $\forall (a,b), (c,d) \in B \times B \ ((a,b) R (c,d) \Leftrightarrow (a \leq c) \wedge (b \leq d)).$

(a) Prove that R is a partial order.

1. **(Reflexivity)** Take any $(a,b) \in B \times B$,

1.1. $a \leq a$ and $b \leq b$.

Aim: 1.2. So $(a,b) R (a,b)$ **by the definition of R .**

1.3. Hence R is reflexive.

2. **(Antisymmetry)** Take any $(a,b), (c,d) \in B \times B$,

2.1. Suppose $(a,b) R (c,d)$ and $(c,d) R (a,b)$.

2.2. Then $a \leq c$, $b \leq d$, $c \leq a$ and $d \leq b$ **by the definition of R .**

2.3. Then $a = c$ and $b = d$ **by the antisymmetry of $\leq$.**

Aim: 2.4. So $(a,b) = (c,d)$ **by equality of ordered pairs.**

2.5. Hence R is antisymmetric.

Let R be a relation on a set A .

1. R is **reflexive** iff $\forall x \in A \ (xRx)$.

2. R is **antisymmetric** iff
 $\forall x, y \in A \ (x R y \wedge y R x \Rightarrow x = y).$

Q6. Let $B = \{0,1\}$ and define the binary relation R on $B \times B$ as follows:
 $\forall (a,b), (c,d) \in B \times B \ ((a,b) R (c,d) \Leftrightarrow (a \leq c) \wedge (b \leq d)).$

(a) Prove that R is a partial order.

3. **(Transitivity)** Take any $(a,b), (c,d), (e,f) \in B \times B$,

3.1. Suppose $(a,b) R (c,d)$ and $(c,d) R (e,f)$.

3.2. Then $a \leq c$, $b \leq d$, $c \leq e$ and $d \leq f$ by the definition of R .

3.3. Then $a \leq e$ and $b \leq f$ by the transitivity of $\leq$.

Aim: 3.4. So $(a,b) R (e,f)$ by the definition of R .

3.5. Hence R is transitive.

4. Therefore R on $B \times B$ is a partial order.

Let R be a relation on a set A .

1. R is **reflexive** iff $\forall x \in A \ (xRx)$.
2. R is **antisymmetric** iff $\forall x, y \in A \ (x R y \wedge y R x \Rightarrow x = y)$.
3. R is **transitive** iff $\forall x, y, z \in A \ (xRy \wedge yRz \Rightarrow xRz)$.

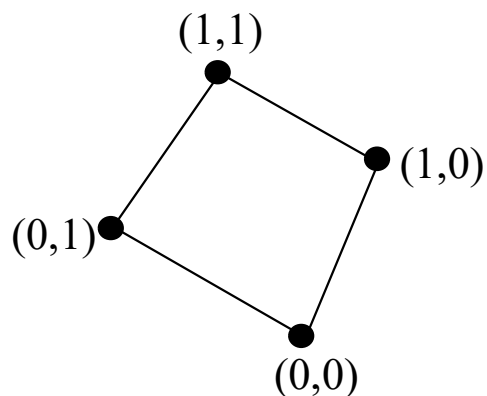
Q6. Let $B = \{0,1\}$ and define the binary relation R on $B \times B$ as follows:

$$\forall (a,b), (c,d) \in B \times B \left((a,b) R (c,d) \Leftrightarrow (a \leq c) \wedge (b \leq d) \right).$$

(b) Draw the Hasse diagram for R .

(c) Find the maximal, largest, minimal and smallest elements.

(d) Is $(B \times B, R)$ well-ordered?



Maximal: (1,1)

Largest: (1,1)

Minimal: (0,0)

Smallest: (0,0)

Not well-ordered.

Reason: It is not even a total order, as (0,1) and (1,0) are not comparable.

Let a set A be partially ordered with respect to a relation $\leq$ and $c \in A$.

1. c is a **maximal element** of A iff $\forall x \in A (c \leq x \Rightarrow c = x)$.
2. c is a **minimal element** of A iff $\forall x \in A (x \leq c \Rightarrow c = x)$.
3. c is the **largest element** of A iff $\forall x \in A (x \leq c)$.
4. c is the **smallest element** of A iff $\forall x \in A (c \leq x)$.

A relation R on a set A is a **total order** iff R is a partial order and $\forall x, y \in A (x R y \vee y R x)$.

Q7.

Let R be a binary relation on a non-empty set A . Let $x, y \in A$. Define a relation S on A by: $x S y \Leftrightarrow (x = y) \vee (x R y)$ for all $x, y \in A$. Show that (a) S is reflexive; (b) $R \subseteq S$; and (c) if S' is another reflexive relation on A and $R \subseteq S'$, then $S \subseteq S'$. What is this relation S called?

(a)

1. Let $x \in A$.
2. $x = x$, so $x S x$ by the definition of S .
3. Therefore, S is reflexive.

Let R be a relation on a set A .
 R is **reflexive** iff $\forall x \in A (xRx)$.



Q7.

Let R be a binary relation on a non-empty set A . Let $x, y \in A$. Define a relation S on A by: $x S y \Leftrightarrow (x = y) \vee (x R y)$ for all $x, y \in A$. Show that (a) S is reflexive; (b) $R \subseteq S$; and (c) if S' is another reflexive relation on A and $R \subseteq S'$, then $S \subseteq S'$. What is this relation S called?

(b)

1. Suppose $(x, y) \in R$, that is, $x R y$.
2. So $x S y$ by the definition of S .
3. So $(x, y) \in S$.
4. Therefore, $R \subseteq S$ by the definition of $\subseteq$.

Aim: To arrive at $(x, y) \in S$.

Q7.

Let R be a binary relation on a non-empty set A . Let $x, y \in A$. Define a relation S on A by: $x S y \Leftrightarrow x = y \vee x R y$ for all $x, y \in A$. Show that (a) S is reflexive; (b) $R \subseteq S$; and (c) if S' is another reflexive relation on A and $R \subseteq S'$, then $S \subseteq S'$. What is this relation S called?

- (c)
1. Suppose $(x, y) \in S$.
 2. Then $x S y$, which means $x = y \vee x R y$ by the defn of S .
 3. Case 1: $x = y$
 - 3.1. Then $x S' y$ since S' is reflexive.
 - 3.2. So $(x, y) \in S'$.
 4. Case 2: $x R y$
 - 4.1. Then $(x, y) \in R \subseteq S'$.
 - 4.2. Then $(x, y) \in S'$.
 5. In all cases, $(x, y) \in S'$.
 6. Therefore, $S \subseteq S'$.

Aim:

S is called the **reflexive closure** of R .
It is the smallest relation on A that is reflexive and contains R as a subset.

Q8. Let R be a binary relation on a set A .

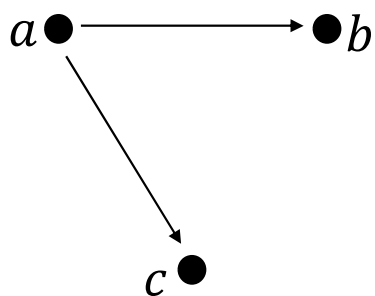
- R is **antisymmetric** iff $\forall x, y \in A (x R y \wedge y R x \Rightarrow x = y)$.
- R is **asymmetric** iff $\forall x, y \in A (x R y \Rightarrow y \not R x)$.

Find a binary relation on A that is ...

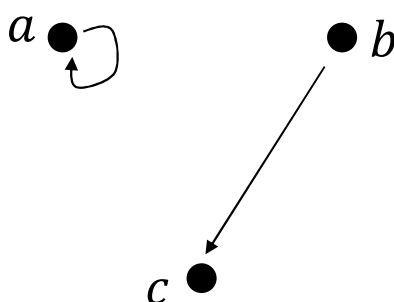
- (a) both asymmetric and antisymmetric.
- (b) not asymmetric but antisymmetric.
- (c) asymmetric but not antisymmetric.
- (d) neither asymmetric nor antisymmetric.

Let $A = \{a, b, c\}$.

(a) $R = \{(a, b), (a, c)\}$.



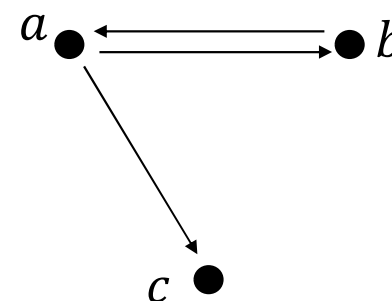
(b) $R = \{(a, a), (b, c)\}$.



(c)

?

(d) $R = \{(a, b), (b, a), (a, c)\}$.



Q8. Let R be a binary relation on a set A .

- R is **antisymmetric** iff $\forall x, y \in A (x R y \wedge y R x \Rightarrow x = y)$.
- R is **asymmetric** iff $\forall x, y \in A (x R y \Rightarrow y \not R x)$.



Find a binary relation on A that is ...

(c) asymmetric but not antisymmetric.

To prove:

Every asymmetric relation
is antisymmetric.

1. Take any binary relation R on a set A .
2. Suppose R is asymmetric.,
 - 2.1. Then $\forall x, y \in A (x R y \Rightarrow y \not R x)$ by the definition of asymmetry.
 - 2.2. $\equiv \forall x, y \in A (x \not R y \vee y \not R x)$ by the implication law.
 - 2.3. $\Rightarrow \forall x, y \in A ((x \not R y \vee y \not R x) \vee x = y)$ by generalization.
 - 2.4. $\equiv \forall x, y \in A (\sim(x R y \wedge y R x) \vee x = y)$ by the De Morgan's law.
 - 2.5. $\equiv \forall x, y \in A (x R y \wedge y R x \Rightarrow x = y)$ by the implication law.
3. Line 2.5 is the definition of antisymmetry, hence R is antisymmetric.

Aim:

Q8. Let R be a binary relation on a set A .

- R is **antisymmetric** iff $\forall x, y \in A (x R y \wedge y R x \Rightarrow x = y)$.
- R is **asymmetric** iff $\forall x, y \in A (x R y \Rightarrow y \not R x)$.



Does this mean that asymmetry $\Leftrightarrow$ antisymmetry and not reflexive?

No!

Take into consideration $A = \{1, 2\}$ and $R = \{(1, 2), (2, 2)\}$.

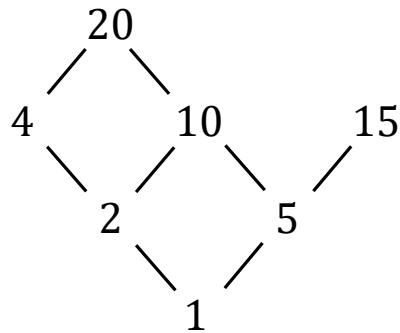
R is not reflexive.

R is antisymmetric.

But R is NOT asymmetric.

So, is there an equivalent statement? Look out for the term irreflexive.

Q9. Consider the “divides” relation on $A = \{1, 2, 4, 5, 10, 15, 20\}$.
List out the pairs of distinct elements in A that are
(a) comparable; (b) compatible.



Let $\leq$ be a partial order on a set A , and $a, b \in A$.

- a, b are **comparable** if $a \leq b$ or $b \leq a$.
- a, b are **compatible** if there is $c \in A$ such that $a \leq c$ and $b \leq c$.

(a) Comparable: $\{1,2\}, \{1,5\}, \{1,4\}, \{1,10\}, \{1,15\}, \{1,20\},$
 $\{2,4\}, \{2,10\}, \{2,20\}, \{5,10\}, \{5,15\}, \{5,20\}, \{4,20\}, \{10,20\}.$

(b) Compatible: $\{1,2\}, \{1,5\}, \{1,4\}, \{1,10\}, \{1,15\}, \{1,20\},$
 $\{2,4\}, \{2,5\}, \{2,10\}, \{2,20\}, \{5,4\}, \{5,10\}, \{5,15\}, \{5,20\},$
 $\{4,10\}, \{4,20\}, \{10,20\}.$

[AY2022/23 Semester 2 Mid-term Test]

Q10.

Let $\leq$ be a partial order on a non-empty set A .

A subset C of A is called a **chain** if and only if every pair of elements in C is comparable, that is, $\forall a, b \in C (a \leq b \vee b \leq a)$. A **maximal chain** is a chain M such that $t \notin M \Rightarrow M \cup \{t\}$ is not a chain. The length of a chain is one less than the number of elements in it.

- (a) Let $A = \{a, b, c, d\}$ and $(\mathcal{P}(A), \subseteq)$ be a poset on $\mathcal{P}(A)$, where $\mathcal{P}(A)$ denotes the power set of A . Write out two maximal chains in $(\mathcal{P}(A), \subseteq)$.

Many possible answers.

$\{ \emptyset, \{a\}, \{a, b\}, \{a, b, c\}, \{a, b, c, d\} \}$

$\{ \emptyset, \{b\}, \{b, d\}, \{b, c, d\}, \{a, b, c, d\} \}$



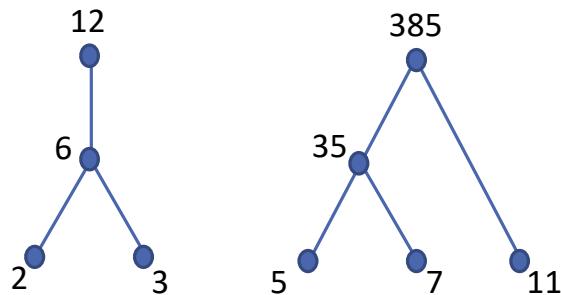
[AY2022/23 Semester 2 Mid-term Test]

Q10.

Let $\leq$ be a partial order on a non-empty set A .

A subset C of A is called a **chain** if and only if every pair of elements in C is comparable, that is, $\forall a, b \in C (a \leq b \vee b \leq a)$. A **maximal chain** is a chain M such that $t \notin M \Rightarrow M \cup \{t\}$ is not a chain. The length of a chain is one less than the number of elements in it.

- (b) Let $B = \{2, 3, 5, 6, 7, 11, 12, 35, 385\}$ and $(B, |)$ be a poset on B , where $|$ denotes the divides relation. Draw the Hasse diagram and write out two maximal chains of different lengths in $(B, |)$.



$\{2, 6, 12\}$ and $\{11, 385\}$

$\{3, 6, 12\}$ and $\{11, 385\}$

$\{5, 35, 385\}$ and $\{11, 385\}$

$\{7, 35, 385\}$ and $\{11, 385\}$

Q11. State whether the following is true or false and justify your answer.

Let $\preceq$ be a partial order on a set A , and $a, b \in A$.

- a, b are *comparable* if $a \preceq b$ or $b \preceq a$.
- a, b are *compatible* if there is $c \in A$ such that $a \preceq c$ and $b \preceq c$.

(a) In all partially ordered set, any two comparable elements are compatible.

True

The 2 cases are symmetrical, so may just use WLOG to prove one of the cases.

1. Let $a, b \in A$ such that a, b are comparable.
2. Then either $a \preceq b$ or $b \preceq a$ by the definition of comparability.
3. Case 1: $a \preceq b$.
 - 3.1. Let $c = b$.
 - 3.2. Then $a \preceq b = c$ by assumption and $b \preceq b = c$ by reflexivity.
 - 3.3. So a, b are compatible by the definition of compatibility.
4. Case 2: $b \preceq a$.
 - 4.1. Let $c = a$.
 - 4.2. Then $b \preceq a = c$ by assumption and $a \preceq a = c$ by reflexivity.
 - 4.3. Hence a and b are compatible by the definition of compatibility.
5. In all cases, a and b are compatible.



Q11. State whether the following is true or false and justify your answer.

Let $\preceq$ be a partial order on a set A , and $a, b \in A$.

- a, b are **comparable** if $a \preceq b$ or $b \preceq a$.
- a, b are **compatible** if there is $c \in A$ such that $a \preceq c$ and $b \preceq c$.

(a) In all partially ordered set, any two comparable elements are compatible.

True

1. Let $a, b \in A$ such that a, b are comparable.
2. Then either $a \preceq b$ or $b \preceq a$ **by the definition of comparability.**
3. WLOG, let $a \preceq b$.
 - 3.1. Let $c = b$.
 - 3.2. Then $a \preceq b = c$ **by assumption and**
 $b \preceq b = c$ **by reflexivity.**
 - 3.3. Hence a and b are compatible **by the definition of compatibility.**
4. Hence a and b are compatible.



Q11. State whether the following is true or false and justify your answer.

Let $\preceq$ be a partial order on a set A , and $a, b \in A$.

- a, b are *comparable* if $a \preceq b$ or $b \preceq a$.
- a, b are *compatible* if there is $c \in A$ such that $a \preceq c$ and $b \preceq c$.

(b) In all partially ordered set, any two compatible elements are comparable.

False

1. Consider the “divides” relation $|$ on $\mathbb{Z}^+$, which is a partial order.
2. Then 2 and 3 are compatible as $2 | 6$ and $3 | 6$.
3. But 2 and 3 are not comparable as $2 \nmid 3$ and $3 \nmid 2$.



End of Tutorial 5

Good luck for your midterms!